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AI-Based Plagiarism Detection System

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Abstract

It has become highly important, alongside the swift proliferation of virtual material and web-based sources, to detect plagiarism. The traditional tools of the plagiarism detection, such as string matching, and fingerprinting are not very effective in locating plagiarized or semantically similar work. The plagiarism detection technologies described in this survey paper overlooked the classical technologies and explained the modern technologies which were developed according to the concepts of Artificial Intelligence (AI) and Natural Language Processing (NLP).

Early systems have been discussed in the paper i.e. the MOSS to identify the plagiarism in codes and the classical text-based systems i.e. stylometry and statistical analysis. It also covers the current practice of employing word embeddings like GloVe and deep learning application to semantically close and pragmatic. These systems that have been enhanced by AI significantly contribute to the accuracy and efficiency of plagiarism detection in identifying the sophisticated patterns of plagiarism besides direct textual matching.

Finally, the survey identifies the primary opportunities in the field, such as detecting paraphrased sources and processing a large amount of information, and outlines the potential directions of the field in the future, such as cross-language plagiarism detection and real-time solutions. The research will attempt to give a clear insight into the Curren Plagiarism Detection methods and their usefulness in contemporary plagiarism detection mechanisms.

Key Words: Plagiarism Detection, Artificial Intelligence, Natural Language Processing, Deep Learning, Word Embeddings, Glove, Semantic Similarity, Text Analysis, Machine Learning, MOSS (Measure of Software Similarity)

I. INTRODUCTION

Due to the fast development of digital technologies and the availability of internet information, plagiarism has become a significant problem in the academic, research, and professional world. The students and the researchers to obtain the content of diverse sources easily, and it is hard to verify the originality using the traditional methods of manual checks through the use of manual checking. Consequently, there is increased demands of effective and automated detection systems of plagiarism.

Both the old techniques of plagiarism detection were merely founded on the matching and comparison of the keys. The methods are fast and easy to perform but do not perform well in isolating more complex cases of plagiarism such as paraphrasing, ideas copying or altering the arrangement. It has caused the creation of more sophisticated methods based on Artificial Intelligence (AI) and Natural Language Processing (NLP).

The new forms of detection of plagiarism are algorithmic in nature in order to replicate not only the superficiality of the text, but the unconscious meaning and context as well. Some of the methods they employ in establishing the semantic similarities of documents include word embeddings, machine learning models and deep learning algorithms. Plagiarism has also been determined in the programming assignment through the use of tools like MOSS that could be used to view the code structure and logic of the programming assignment.

This survey paper aims to provide a general picture of the list of potential methods of detection of plagiarism starting with the traditional methods to the most effective AI methods. It also signifies the strengths, weaknesses and problems of each method and also the chances of the future perspective of the methods of detecting plagiarism through futuring means.

II. RELATED WORK

A. Evolution of Plagiarism Detection

Plagiarism detection systems dating back prior to modern times typically could only do so based upon comparing strings of text. The detection systems could identify when the same (exact) words were contained within 2 pieces of writing; therefore, they did not work well for things like "paraphrased", modified or changed content. After some advances were made in technology, systems for detecting plagiarism based upon different methods such as fingerprinting, citation analysis, and stylometry were developed and implemented resulting in an improvement over comparison of strings of text. However, even with these advanced techniques, systems are still limited in their ability to link the meaning of the text.

B. Types of Plagiarism:

It is crucial to understand the types of plagiarism to help develop effective plagiarism-detection systems. The four most common types of plagiarism include:

(1) Direct Plagiarism - copying the text word-for-word;

(2) Paraphrased Plagiarism - changing the text with only a few words unchanged;

(3) Self-plagiarism - reusing one's previous work;

(4) Mosaic Plagiarism - copying and mixing the text with your own original ideas. In addition to these four major types of plagiarism, there is also an eighth type of plagiarism known as Idea Plagiarism.

C. Existing Plagiarism Detection Tools:

Several tools are currently used in academia and industry. A comparison is shown below:

Table 1: Comparison of Existing Tools

Tool	Technique used	Strengths	Limitations
Turnitin	Database Matching	Large database	Weak in paraphrasing
Grammarly	AI + Grammar Check	Easy to use	Limited plagiarism depth
Copyscape	Web Matching	Good for web content	Not suitable for academics
Quetext	Deep Search	Contextbased detection	Limited database

These tools are useful but still struggle with detecting meaning-based similarities.

III. AI TECHNIQUES USED IN PLAGIARISM DETECTION

A. Natural Language Processing (NLP)

Understanding how people talk gets easier because machines learn patterns. Sentence shapes, word rules, maybe even intent those get broken down piece by piece. Chopping words into parts, reducing them, digging into what they actually mean these steps show up a lot.

B. Machine Learning (ML)

Machine Learning models are trained on datasets so they can identify patterns in text. These models can classify if a piece of content is plagiarized or not based on the learned features.

C. Deep Learning Approaches

Deep learning models like LSTM and transformers (such as BERT) are way more advanced and can capture context and meaning more effectively. These models are useful for detecting paraphrased plagiarism.

Table 2: AI Techniques Comparison

Technique	Advantage	Limitation
NLP	Understand structure	Limited deep context
ML	Learn patterns	Needs training data

Deep learning	Capture meanings	High computation cost

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed system is designed to overcome limitations of outdated tools by focusing on semantic understanding.

A. System Overview

The plagiarism detection system under consideration is aimed at enhancing the current systems in terms of text matching and meaning analysis. Specifically, the system is built in several layers, each responsible for a certain operation within the process. First of all, there is an input layer, which implies uploading a text in one of the following formats: PDF or DOC. Then the preprocessing layer performs certain actions with respect to the text; these include deleting unwanted symbols, transforming the text to a standardized form, and splitting it into sentences and words.

Once it is done, the feature extraction layer translates the modified text into numeric form. After that, the AI analysis layer conducts the comparison of the obtained data against already available documents on the basis of semantic approach. Eventually, the output layer creates a report containing information about the level of similarity, highlighted text fragments, and other details.

B. Working Process

The operation process of the suggested system comprises several stages that help detect plagiarism effectively. First, the user submits the file via the interface of the program. After submitting, the file is sent to the pre-processing phase, where all unnecessary elements such as stop words, punctuation, and formatting issues are removed from the text. Further, the text is split into tokens and transformed into vectors.

The next phase is featuring extraction and conversion of the data into vectors, which is followed by their comparison to other documents stored in a database using an AI algorithm. The system analyzes not only identical copies of the text but also semantically similar texts. In the end, the program provides a percentage of the similarity between the submitted document and others as well as recommendations for improvement.

C. Features of the Proposed System

The suggested system has various attributes that enhance its efficacy when compared to existing plagiarism detection methods. One of the attributes is the capacity of the suggested system to detect paraphrasing in the documents. The proposed system understands the semantic of the sentences and not only searches for similar words, thus enabling the identification of sophisticated plagiarism cases. Other important attributes include the provision of AI-powered feedback to users, which recommends ways to enhance their content and ensure low similarity. Furthermore, the system offers a user-friendly interface that shows results through a dash board indicating similarity percentage and highlighted text. It also supports different document types and large file sizes. The system can support many users simultaneously, making it scalable.

V. RESULTS AND PERFORMANCE EVALUATION

The AI-based plagiarism detection system we proposed has provided considerable betterment against conventional systems. Unlike older models that merely relied on straight pattern matching to find possible instances of plagiarism, this proposed system was developed to attempt to understand the meaning of the text in order to identify paraphrasing and semantic similarities. With this ability comes greater accuracy level. In addition, the level of false-positive results is reduced by eliminating many of the instances where common phrases are flagged inaccurately in older systems. It's justified by the improved quality of detection provided by this AI-based model over previous generations. The overall performance will provide effective results when identifying both direct and indirect instances of plagiarism.

Table 3: Expected Performance Comparison

System type	Accuracy	Paraphrased detection	Speed
Traditional tool	Medium	Low	Fast
Ai based system	High	High	Moderate

VI. LIMITATIONS AND FUTURE WORKS

While the system has many advantages, it also has some limitations. It requires large data sets for training, computational power, and an extended timeframe in order to operate. The concept-based plagiarism detection has proven to be difficult to detect. In the future, the system could be improved through optimizing the efficiency of the model and reducing its computational costs. The system could also be made user friendly and real-world use if it supported a lot other languages and help with real-time detection methods.

VII. CONCLUSION

In summary, the transition from conventional, non-AI plagiarism detection software to AI-based plagiarism detection software is a significant development in evaluating originality in written material. While original process did rely heavily on exact word matching (the previous method), it was unable to detect instances of paraphrased or slightly altered versions of the original work. The new system will use artificial intelligence and natural language processing technology, which allows it to analyze the meaning of text and its context, so it can find deeper similarities between two pieces of text, even when they have been marked with different words.

Another benefit of this system is that it will give users a greater level of accuracy and detail in their final reports. Users will receive a percentage of how much their work is plagiarized, and they will receive valuable information regarding which specific portions of their work are at risk of being plagiarized. As a result of this, users, including students, researchers, and professionals who use this software will not only be alerted that they have copied from someone else, but they will also have the opportunity to fix their work and continue developing superior quality work.

The plagiarism detection system is not perfect, however. It is still a heavily reliant upon large data sets and computing capabilities, and identifying idea patterns of plagiarizing is still very challenging. Nevertheless, improvements in AI will provide for more effective and faster systems in the near future.

Ultimately, AI-based plagiarism detection software is important to promoting academic integrity and to providing means to create authentic material.

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